

AI-Augmented Research

Session 3.2: Research Judgement in an Institutional Context

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instats Seminar

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Interactive App and Repository

- **Streamlit App URL:** [Click here to open the Interactive Streamlit App](#)
- **No Installation:** Runs entirely in the browser via Streamlit — no Python or programming knowledge required.
- **Navigation:** Use the sidebar to switch between the six seminar sessions across the three days.
- **Step-by-Step:** Each page walks through worked examples and a Bring Your Own Data (BYOD) interface for that session's core workflows — from research question development and literature scoping to data quality auditing, end-to-end computational pipelines, manuscript integrity checks, and AI disclosure evaluation.
- **Consistency:** Results match the companion Python scripts and prompt library available in the `scripts/` folder of the seminar GitHub repository.
- **GitHub Page URL:** [Click here to open the seminar GitHub page](#)

Session 3.2 Overview

- 1 The Institutional Policy Landscape
- 2 Publisher and Funder Policies
- 3 Reproducibility and Transparency
- 4 Future Directions
- 5 Closing Synthesis

The Policy Landscape Is Underdeveloped Relative to the Technology

The rapid adoption of AI in research has outpaced the development of institutional policies. The most common failure: policies that address plagiarism and disclosure but do not address the **validity and reproducibility challenges** that AI use creates.

- Zyphur (2026): only **6 of 38** top-tier doctoral universities have AI policies that extend past plagiarism into AI literacy and valid research practices
- Blau et al. (2024, *PNAS*): institutional oversight as one of the five principles of scientific integrity in an AI age

This means researchers cannot rely on their institutions to provide the AI literacy framework they need. That is what this seminar is for.

The Zyphur Institutional Benchmarking Grid (Dimension 5)

Zyphur (2026) proposes a four-axis benchmarking grid for assessing institutional AI readiness:

Axis	Description	What a strong policy includes
Policy document	The written AI policy	Extends past plagiarism to validity and reproducibility
Named responsibility	People accountable for enacting the policy	Research integrity officer, AI literacy coordinator
Technology systems	Tools that support responsible AI use	Institutional RAG systems, approved tool lists
Process integration	AI use in supervision and examination	PhD supervision protocols, thesis examination criteria

Most institutions are strong on the first axis and weak on the other three.

Where Doctoral Universities Currently Stand

Zyphur's (2026) audit of 38 top-tier doctoral universities found:

- **38/38** have some form of AI policy document
- **32/38** policies focus primarily on plagiarism and disclosure
- **6/38** policies extend to AI literacy and valid research practices
- **4/38** have named individuals responsible for AI literacy in research training
- **2/38** have integrated AI literacy into PhD supervision protocols

This reflects the speed of the technology's adoption. Researchers must develop their own AI literacy framework rather than relying on institutional guidance.

Publisher Policies: A More Developed Landscape

Publishers have moved faster than universities on AI policy, partly because they face direct reputational risk from AI-related integrity failures.

- All major publishers now prohibit listing AI as an author
- Most major publishers require disclosure of AI use in writing and data analysis
- Some publishers (*Nature*, *Science*) have extended policies to cover AI use in peer review
- **The frontier:** policies on AI use in data collection, annotation, and statistical analysis are still underdeveloped

Nature (2026): hallucinated citations are polluting the scientific literature — the policy response is lagging.

Funder Policies: An Emerging Frontier

Research funders are beginning to address AI use in funded research.

Funder	Key policy	Disclosure requirement
NIH (2024)	Human oversight required	Yes — in grant applications and reports
NSF (2024)	AI use must be disclosed	Yes — in applications and progress reports
ERC (2024)	Must comply with GDPR and research integrity	Yes
Wellcome Trust (2025)	AI cannot be listed as author	Yes — in research outputs

Common thread: disclosure and human accountability. The frontier: validity and reproducibility requirements.

Reproducibility and Transparency in AI-Mediated Research

Reproducibility and transparency take on new meanings in AI-mediated research. The challenge: LLMs are non-deterministic, model versions change, and the training data is opaque.

- Baltes et al. (2025): non-determinism, opaque training data, and rapidly evolving models threaten reproducibility and replicability
- Rover et al. (2026): non-determinism persists and amplifies in multi-turn interactions

A reproducibility standard for AI-mediated research must address all three challenges: non-determinism, model version change, and opaque training data.

A Proposed Reproducibility Standard for AI-Mediated Research

Drawing on Zyphur (2026) Competency 2 and the emerging literature:

- ① **Model specification:** name, version, provider, and date of access
- ② **Parameter specification:** temperature, top-p, seed (where available)
- ③ **Prompt documentation:** full prompt text, verbatim
- ④ **Output documentation:** full output or representative sample
- ⑤ **Re-run stability test:** run the same prompt 3–5 times; report whether outputs are substantively identical
- ⑥ **Fallback documentation:** if the specific version is no longer available, document it and provide the full prompt for attempted replication

The Transparency Obligation: Reporting AI Use in Methods Sections

Template for reporting AI use in a methods section

“[Step name] was performed with the assistance of [AI tool name, version, date of access]. The prompt used was: ‘[exact prompt text]’. The model was run at temperature [T] with seed [S]. The output was verified by [verification method]; [N] errors were identified and resolved as follows: [description]. The full output is available as Supplementary Material [X].”

This template satisfies the documentation requirements of Zyphur (2026) Competency 2 and the disclosure requirements of major publishers.

The Structured Failure-Mode Report in Practice

Template for the structured failure-mode report (Zyphur Competency 6)

Protocol: [AI tool, version, prompt, parameters, date].

Threat model: [specific failure modes relevant to this type of output — hallucination, non-determinism, sycophancy, coverage gaps].

Failure-discovery rate: $[N \text{ failures} / M \text{ outputs checked} = X\%]$.

Failure resolution: [how failures were identified and resolved].

Residual risk: [what cannot be fully verified and why].

This report is not a disclosure of weakness; it is a demonstration of methodological rigor.

Future Directions: Three Developments That Will Reshape Research

- ➊ **Agentic research systems:** AI agents that can autonomously execute multi-step research workflows, from literature discovery to data collection to analysis
- ➋ **AI-assisted peer review at scale:** AI systems that can screen manuscripts for methodological concerns, statistical errors, and citation validity before human review
- ➌ **AI-generated hypotheses:** AI systems that can propose novel research questions by identifying gaps and contradictions in the existing literature

Each development **amplifies both the potential and the risks** of AI-augmented research.

Agentic Research Systems: The Frontier

AI agents that can autonomously execute multi-step research workflows are already in early deployment. Current examples: Elicit's automated systematic review pipeline, OpenAI's Deep Research, Perplexity's research mode.

Verification challenge:

Each autonomous step introduces a potential error the researcher may not see.

Reproducibility challenge:

Agent behavior is particularly sensitive to model updates; the full prompt history of a multi-step interaction may be very long.

Nature (2025): AI agents are increasingly used for complex, multi-step research processes.

Human-AI Collaboration in Scientific Discovery: A Calibrated View

The most productive framing is not “AI will replace researchers” or “AI is just a tool.” It is:

AI and human researchers have **complementary strengths** that, when properly combined, produce better research than either could produce alone.

AI's strengths	Human researchers' strengths
Speed and breadth	Domain expertise
Pattern recognition across large corpora	Theoretical judgement
Tireless consistency	Ethical reasoning
Language generation	Interpretive authority
—	Accountability

The Competency Framework as a Durable Foundation

Zyphur (2026) argues that AI literacy is not a one-time tooling skill but an **ongoing exercise of scholarly judgement** that must be built into research training at every level.

The six core competencies are durable because they address the **fundamental challenges** of AI-mediated research, not the specific capabilities of current tools:

- 1 Citation verification
- 2 Model and parameter specification
- 3 Prompt discipline
- 4 Model-heterogeneity in adversarial review
- 5 Sycophancy detection and human-as-verifier discipline
- 6 Structured failure-mode reporting

As tools change, the competencies remain relevant.

The Seminar's Three-Day Arc

The three days of this seminar trace a complete arc:

- ① **Day 1:** AI and the research lifecycle — establishing the conceptual foundations and the verification disciplines for early-stage research
- ② **Day 2:** AI and computational workflows — applying the verification disciplines to data work, demonstrated through a complete end-to-end case study
- ③ **Day 3:** AI and scholarly communication — extending the verification disciplines to writing, peer review, and the institutional context

The organizing thread — **research judgement** — is present throughout.

Applying the Framework After the Seminar

The framework can be applied to any AI tool or workflow the researcher encounters. Three questions to ask at every AI-assisted step:

① Is this a labour task or a judgement task?

If judgement: the researcher decides; AI may assist but not substitute.

② Is this a discovery activity or a verification activity?

If verification: the researcher checks every AI output against an independent standard.

③ Which of the six competencies applies here?

Apply the relevant competency and document the result.

These three questions are the practical expression of the seminar's organizing thread.

Closing: What the Researcher Must Decide

This seminar ends where it began: with the question of what the researcher must decide, and what AI can safely assist with.

The answer is not a fixed list of tasks. It is a **discipline** — the habit of asking, at every AI-assisted step, whether the researcher has maintained the intellectual authority that makes the research valid.

AI is a powerful tool. Research judgement is what makes it a scholarly one.

Zyphur (2026): AI literacy is an ongoing exercise of scholarly judgement, not a one-time tooling skill.

References

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Questions and Discussion

Thank you!

Questions?

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